1)

Security Attacks, Threats. Risks & Security Goals.

Security Attacks

Threats

Risks

Security Goals.

Malware

unauthorized access

Data Breach (Finacial loss)

Confidentiality

Phishing

Steal Information

Identity theft

Integrity

MITM

Data Interception

Data tamping

Availability (ensures data is not intercepte

DOS

Service disruption

Loss of Service

2)

Substitution Chiphers

Ceaser Cipher	Monoalphabetic	Vigenere	Hill
key usage Shift value (0-25)	key Usage 26-letter Substitution	key usage Keyword (Multiple Shifts)	key Usage Key Matric (eg. 2x2, 3x3)
Complexcity Very Low	complexcity Low	Complexcity. medium	complexcity. High.
Security Strength Weak	Security Strength weak	Security Strength Moderate	Security Strength Strongest.

3) CLASSICAL ENCRYPTION SCHEMES.

	Substitution	Transposition
	(Replace characters	(rearrange characters)
	Eg Caser, Vignere, Hill	eg, Rail Fence, Columi
	A → D	Row Transposition A B C D. A C B D
Data Structure	changes the actual characters (same length)	changes the order (Same characters same length)
Security level	Generally Weaker	Slight Stronger (less affected by Frequency)
Vulnerability	Easier to boreak (Freavency analysis)	More resistant (but vulnerable to pattern analysis)

4)

Steganography

Data Hiding

Hides Secret data inside a cover File (image, audio Video txt)

Example:

Hidden txt in an image

Water Marking

Adds Owner Ship info to content

Example

Digital water mark in Photo/videos

Survivability

Survives attacks/ Changes (Noise Compression, etc)

Example

Robust water marking in media.

Together, they help in Scure communicat

by hiding, Protesting & kepping the data Safe